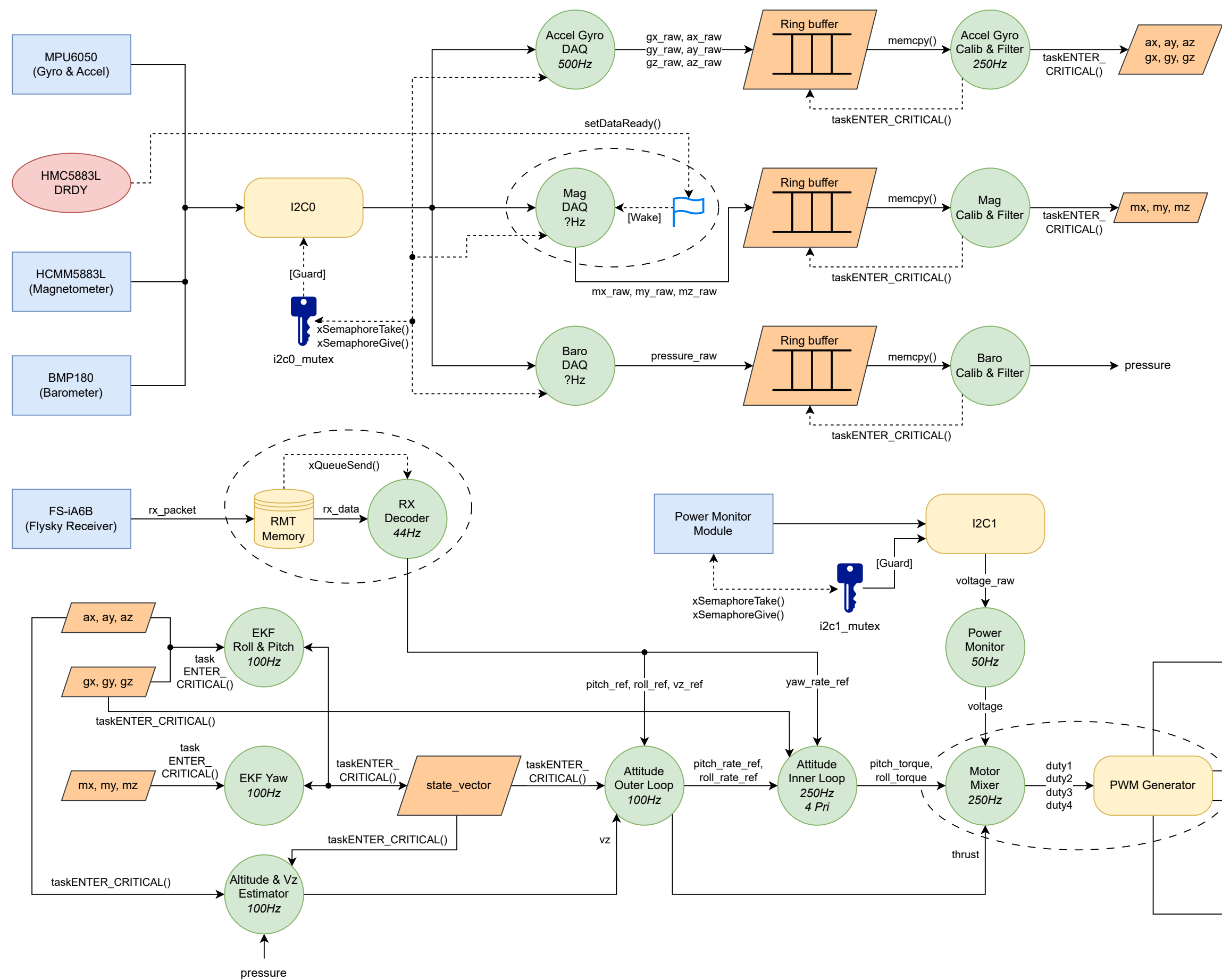


FLIGHT CONTROL SYSTEM
RTOS ARCHITECTURE

Designer: Nguyen Quy Hao Nhlen



Conventions

- **Green Circle:** RTOS tasks, where CPUs will directly spend there cycles doing calculations
- **Yellow Rounded Rectangle:** SoC's independent hardware peripherals, stand alone from CPU, doesn't interfere with CPU cycles
- **Blue Rectangle:** External hardware
- **Red Ellipse:** Interrupt Service Routines (ISR)
- **Orange Parallelogram:** Shared memory, memory chunk, variables. Protection methods must exist
- Solid arrow: Data flow
- Dashed arrow: Control & Synchronization Signals
- Key: mutex
- Horizontal Ladder: Queue / Ring buffer
- Flag: flagging variable (boolean)

Notes & Rules

- Storing ring buffer data must also store its newest index